

Computer Networks 2020-2021

Spring Midterm Exam

80 minutes

First two questions will be answered as text. Other questions will be solved to paper, scanned and uploaded; separately for each question.



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1

(10 pts)

- Guess the most widely used network protocol.
- Provide a strict reasoning for your answer.

IP,TCP

Every device connected to the internet has an IP address and these devices communicate via IP address. Because the data loss integrity is important in most communications, TCP is used

2

(16 pts) Which is used in Internet: Packet or circuit switching? Write two important reasons.

Packet Switching

- There is no need to dedicate a channel for the selected data transmission. That allows more data flow in time unit, thus it is more cost effective.
- the data packets that we send can find the route of a given destination by itself.

3

(10 pts) There is a connection where distance between sender and receiver is 1000 km. Calculate the RTT by assuming that propagation speed is 2×10^5 km/s. Neglect the queue and processing delay. (Non-anonymous question ⓘ)

 3.jpg

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4

(24 pts)

a) We want to transfer a file of size 10^4 bytes. Assume that there is a 40 bytes header for each segment and maximum segment size is 1500 bytes. How many segments does it need at least to transfer this file? Calculate the transfer time for each segment by assuming the connection is 5 Mbps.

b) Assume that we use Selective Repeat algorithm with window size $N=6$ to transfer the file from (a). The losses occur where data segment 4 and ACK segment 2 are sent on their first transmission. All the data and ACK segments except these 2 segments are sending to receiver without loss. Timeout for each segment is 10 ms and is started after the segment is sent. How long does it take to transfer whole file and to receive last ACK for receiver? Draw the timeline. (Non-anonymous question ⓘ)

 4.jpg

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5

(20 pts) You are asked to make subnet addressings of 4 buildings in your company. Below are the PC numbers for each building. Assume that the 200.55.180.0/18 address block is reserved for your company.

a) Specify the addresses of the 4 subnets you created in the shown table. Show your work.

b) What is the size of the largest single CIDR address block that you can assign from the unassigned addresses in the address block 200.55.180.0/18 remaining after you assigned the addresses to these four buildings?

(Alternative link for the image:

<https://drive.google.com/file/d/1dJvSqcqBC1wYqzg3fuWa0pPiBHMwzN9/view?usp=sharing>) (Non-anonymous question ⓘ)

Building	# PC	Subnet Address
1	1605	
2	4000	
3	1022	
4	1024	

 5.jpg

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6

(20 pts) According to the shown routing table, which output ports are IP packets with the shown destination addresses routed to? Show your work in detail.

(Alternative link for the image:

<https://drive.google.com/file/d/1gDCjGU26l58MxY1TM3aeWKHUKD58DU45/view?usp=sharing>) (Non-anonymous question ⓘ)

Target Network	Next Hop
222.88.144.0/19	R3
222.88.0.0/16	R4
222.88.128.0/18	R1
222.88.160.0/21	R5
222.88.0.0/17	R2

- De:**
- I. 222.
 - II. 222.
 - III. 222.
 - IV. 222.
 - V. 222.

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